Initial Parameter Configuration.

```
Dropout = 0.1
SEED = 42
layers = 2
optimizer='adam'
loss='mse'
WINDOW=100
Units=32
Epochs=20
Patience=10
batch_size = 32
5 Year Data (15m)
RMSE (Root Mean Squared Error)
MAE (Mean Absolute Error)
```

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 100, 32)	4,352
dropout (Dropout)	(None, 100, 32)	0
lstm_1 (LSTM)	(None, 32)	8,320
dropout_1 (Dropout)	(None, 32)	0
dense (Dense)	(None, 1)	33

Total params: 12,705 (49.63 KB)
Trainable params: 12,705 (49.63 KB)
Non-trainable params: 0 (0.00 B)

learning_rate = 0.0001

```
Epoch 1/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 28s 11ms/step - loss:
2.5998e-04 - val_loss: 4.9809e-05
Epoch 2/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 41s 11ms/step - loss:
2.5141e-04 - val_loss: 5.5730e-05
Epoch 3/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 44s 12ms/step - loss:
2.5183e-04 - val_loss: 5.4192e-05
Epoch 4/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 38s 11ms/step - loss:
2.5092e-04 - val_loss: 4.9640e-05
Epoch 5/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 26s 11ms/step - loss:
2.5137e-04 - val_loss: 5.1118e-05
Epoch 6/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 43s 12ms/step - loss:
2.4815e-04 - val_loss: 5.0565e-05
Epoch 7/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 40s 12ms/step - loss:
2.4649e-04 - val_loss: 5.1106e-05
Epoch 8/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 41s 12ms/step - loss:
2.5015e-04 - val_loss: 4.8119e-05
Epoch 9/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 26s 11ms/step - loss:
2.5146e-04 - val_loss: 4.8853e-05
Epoch 10/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 41s 11ms/step - loss:
2.4642e-04 - val_loss: 4.9630e-05
Epoch 11/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 43s 12ms/step - loss:
2.4640e-04 - val_loss: 5.2822e-05
Epoch 12/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 41s 12ms/step - loss:
2.4772e-04 - val_loss: 5.3823e-05
Epoch 13/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 41s 12ms/step - loss:
2.4954e-04 - val_loss: 5.1360e-05
Epoch 14/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 39s 11ms/step - loss:
2.4928e-04 - val_loss: 5.3013e-05
Epoch 15/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 41s 11ms/step - loss:
2.4785e-04 - val_loss: 5.5267e-05
Epoch 16/20
```

```
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 41s 11ms/step - loss:
2.4697e-04 - val_loss: 4.7736e-05
Epoch 17/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 43s 12ms/step - loss:
2.4599e-04 - val_loss: 5.0175e-05
Epoch 18/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 26s 11ms/step - loss:
2.4581e-04 - val_loss: 4.6335e-05
Epoch 19/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 43s 12ms/step - loss:
2.4342e-04 - val_loss: 4.6866e-05
Epoch 20/20
2325/2325 ━━━━━━━━━━━━━━━━━━━━ 40s 12ms/step - loss:
2.5479e-04 - val_loss: 6.5109e-05
Restoring model weights from the end of the best epoch: 18.

2325/2325 ━━━━━━━━━━━━━━━━━━━━ 11s 5ms/step - loss:
1.2437e-05
MSE Train: 1.3275429409986828e-05
773/773 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - loss:
2.8172e-05
MSE Test: 4.667701432481408e-05
773/773 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - loss:
9.5409e-05
MSE Val: 8.771794819040224e-05

Test Set RMSE (Original Scale): 0.000943 EURUSD
Test Set MAE (Original Scale): 0.000604 EURUSD
Validation Set RMSE (Original Scale): 0.001293 EURUSD
Validation Set MAE (Original Scale): 0.000859 EURUSD
Training Set RMSE (Original Scale): 0.000503 EURUSD
Training Set MAE (Original Scale): 0.000351 EURUSD
```

learning_rate = 0.001, WINDOW 30

```
Epoch 1/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 23s 8ms/step - loss: 0.0090
- val_loss: 1.0563e-04
Epoch 2/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 39s 8ms/step - loss:
9.5649e-04 - val_loss: 9.7825e-05
Epoch 3/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 20s 8ms/step - loss:
4.4550e-04 - val_loss: 1.3305e-04
Epoch 4/20
```

```
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 19s 8ms/step - loss:
3.2026e-04 - val_loss: 1.0750e-04
Epoch 5/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 20s 8ms/step - loss:
2.9688e-04 - val_loss: 9.1012e-05
Epoch 6/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 20s 8ms/step - loss:
2.9771e-04 - val_loss: 5.5013e-05
Epoch 7/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 21s 8ms/step - loss:
2.9964e-04 - val_loss: 7.9895e-05
Epoch 8/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 21s 8ms/step - loss:
2.9558e-04 - val_loss: 8.6627e-05
Epoch 9/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 20s 8ms/step - loss:
2.8727e-04 - val_loss: 7.4607e-05
Epoch 10/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 20s 8ms/step - loss:
2.8777e-04 - val_loss: 1.4993e-04
Epoch 11/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 21s 8ms/step - loss:
2.8402e-04 - val_loss: 5.7577e-05
Epoch 12/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 20s 8ms/step - loss:
2.8628e-04 - val_loss: 6.3736e-05
Epoch 13/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 20s 8ms/step - loss:
2.8138e-04 - val_loss: 5.8699e-05
Epoch 14/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 21s 8ms/step - loss:
2.8750e-04 - val_loss: 1.3693e-04
Epoch 15/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 20s 8ms/step - loss:
2.8466e-04 - val_loss: 4.1343e-05
Epoch 16/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 21s 9ms/step - loss:
2.8076e-04 - val_loss: 5.8082e-05
Epoch 17/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 20s 8ms/step - loss:
2.8173e-04 - val_loss: 5.7100e-05
Epoch 18/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 19s 8ms/step - loss:
2.8269e-04 - val_loss: 4.1274e-05
Epoch 19/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 20s 8ms/step - loss:
2.8045e-04 - val_loss: 7.9445e-05
Epoch 20/20
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 21s 8ms/step - loss:
2.7307e-04 - val_loss: 5.2788e-05
Restoring model weights from the end of the best epoch: 18.
```

```
2327/2327 ━━━━━━━━━━━━━━━━━━━━ 9s 4ms/step - loss:
2.2763e-05
MSE Train: 1.9447197701083496e-05
775/775 ━━━━━━━━━━━━━━━━━━━━ 3s 4ms/step - loss:
2.3816e-05
MSE Test: 4.127405190956779e-05
775/775 ━━━━━━━━━━━━━━━━━━━━ 3s 4ms/step - loss:
8.0359e-05
MSE Val: 7.31315158191137e-05

Test Set RMSE (Original Scale): 0.000887 EURUSD
Test Set MAE (Original Scale): 0.000577 EURUSD
Validation Set RMSE (Original Scale): 0.001181 EURUSD
Validation Set MAE (Original Scale): 0.000796 EURUSD
Training Set RMSE (Original Scale): 0.000609 EURUSD
Training Set MAE (Original Scale): 0.000480 EURUSD
```

learning_rate = 0.001, WINDOW 100 (10 years data)

```
Epoch 1/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 61s 13ms/step - loss:
0.0032 - val_loss: 7.7900e-05
Epoch 2/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 81s 12ms/step - loss:
2.6301e-04 - val_loss: 6.0482e-05
Epoch 3/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 79s 12ms/step - loss:
2.2866e-04 - val_loss: 1.3478e-04
Epoch 4/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 58s 12ms/step - loss:
2.2184e-04 - val_loss: 4.9466e-05
Epoch 5/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 79s 12ms/step - loss:
2.1804e-04 - val_loss: 3.8836e-05
Epoch 6/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 58s 12ms/step - loss:
2.1258e-04 - val_loss: 4.4267e-05
Epoch 7/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 79s 12ms/step - loss:
2.0713e-04 - val_loss: 2.6390e-05
Epoch 8/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 86s 13ms/step - loss:
2.0806e-04 - val_loss: 3.7515e-05
Epoch 9/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 82s 13ms/step - loss:
2.0607e-04 - val_loss: 6.2626e-05
Epoch 10/20
```

```
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 81s 12ms/step - loss:
2.0558e-04 - val_loss: 3.9207e-05
Epoch 11/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 82s 12ms/step - loss:
2.0196e-04 - val_loss: 3.2587e-05
Epoch 12/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 82s 12ms/step - loss:
1.9881e-04 - val_loss: 5.0556e-05
Epoch 13/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 83s 12ms/step - loss:
2.0102e-04 - val_loss: 2.5359e-05
Epoch 14/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 58s 12ms/step - loss:
1.9520e-04 - val_loss: 2.1924e-05
Epoch 15/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 83s 13ms/step - loss:
1.9359e-04 - val_loss: 3.2253e-05
Epoch 16/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 81s 12ms/step - loss:
1.9491e-04 - val_loss: 3.2447e-05
Epoch 17/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 82s 12ms/step - loss:
1.9153e-04 - val_loss: 2.3812e-05
Epoch 18/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 78s 12ms/step - loss:
1.8935e-04 - val_loss: 3.6150e-05
Epoch 19/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 55s 12ms/step - loss:
1.9080e-04 - val_loss: 4.2437e-05
Epoch 20/20
4660/4660 ━━━━━━━━━━━━━━━━━━━━ 58s 12ms/step - loss:
1.8940e-04 - val_loss: 3.4546e-05
Restoring model weights from the end of the best epoch: 14.

4660/4660 ━━━━━━━━━━━━━━━━━━━━ 22s 5ms/step - loss:
1.7484e-05
MSE Train: 1.4966237358748913e-05
1552/1552 ━━━━━━━━━━━━━━━━━━━━ 7s 5ms/step - loss:
1.5966e-05
MSE Test: 2.1923575332039036e-05
1552/1552 ━━━━━━━━━━━━━━━━━━━━ 8s 5ms/step - loss:
3.5728e-05
MSE Val: 1.6397856597905047e-05

Test Set RMSE (Original Scale): 0.000647 EURUSD
Test Set MAE (Original Scale): 0.000424 EURUSD
Validation Set RMSE (Original Scale): 0.000559 EURUSD
Validation Set MAE (Original Scale): 0.000391 EURUSD
Training Set RMSE (Original Scale): 0.000534 EURUSD
Training Set MAE (Original Scale): 0.000373 EURUSD
```

learning_rate = 0.001, WINDOW 30 (10 years data)

```
Epoch 1/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 53s 11ms/step - loss:
0.0029 - val_loss: 1.1054e-04
Epoch 2/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 41s 9ms/step - loss:
2.6767e-04 - val_loss: 4.7705e-05
Epoch 3/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 80s 8ms/step - loss:
2.2885e-04 - val_loss: 3.4333e-05
Epoch 4/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 47s 10ms/step - loss:
2.2228e-04 - val_loss: 5.4315e-05
Epoch 5/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 77s 9ms/step - loss:
2.1336e-04 - val_loss: 3.2840e-05
Epoch 6/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 40s 9ms/step - loss:
2.1259e-04 - val_loss: 4.8025e-05
Epoch 7/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 40s 9ms/step - loss:
2.0768e-04 - val_loss: 6.7665e-05
Epoch 8/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 45s 10ms/step - loss:
2.0700e-04 - val_loss: 3.7872e-05
Epoch 9/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 77s 9ms/step - loss:
2.0313e-04 - val_loss: 3.6511e-05
Epoch 10/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 41s 9ms/step - loss:
2.0123e-04 - val_loss: 3.5558e-05
Epoch 11/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 81s 9ms/step - loss:
2.0209e-04 - val_loss: 4.2400e-05
Epoch 12/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 47s 10ms/step - loss:
1.9689e-04 - val_loss: 3.3490e-05
Epoch 13/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 82s 10ms/step - loss:
1.9807e-04 - val_loss: 4.3631e-05
Epoch 14/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 46s 10ms/step - loss:
1.9644e-04 - val_loss: 2.7151e-05
Epoch 15/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 76s 9ms/step - loss:
1.9722e-04 - val_loss: 2.0773e-05
Epoch 16/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 41s 9ms/step - loss:
1.9215e-04 - val_loss: 1.8315e-05
```

```
Epoch 17/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 41s 9ms/step - loss: 1.9152e-04 - val_loss: 3.1034e-05
Epoch 18/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 41s 9ms/step - loss: 1.9164e-04 - val_loss: 2.3858e-05
Epoch 19/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 87s 10ms/step - loss: 1.9146e-04 - val_loss: 2.7448e-05
Epoch 20/20
4663/4663 ━━━━━━━━━━━━━━━━━━━━ 83s 10ms/step - loss: 1.9261e-04 - val_loss: 2.5796e-05
Restoring model weights from the end of the best epoch: 16.

4663/4663 ━━━━━━━━━━━━━━━━━━━━ 18s 4ms/step - loss: 1.3238e-05
MSE Train: 1.018812235997757e-05
1554/1554 ━━━━━━━━━━━━━━━━━━━━ 6s 4ms/step - loss: 1.3779e-05
MSE Test: 1.8314833141630515e-05
1554/1554 ━━━━━━━━━━━━━━━━━━━━ 6s 4ms/step - loss: 1.4632e-05
MSE Val: 9.387572390551213e-06

Test Set RMSE (Original Scale): 0.000591 EURUSD
Test Set MAE (Original Scale): 0.000378 EURUSD
Validation Set RMSE (Original Scale): 0.000423 EURUSD
Validation Set MAE (Original Scale): 0.000288 EURUSD
Training Set RMSE (Original Scale): 0.000441 EURUSD
Training Set MAE (Original Scale): 0.000296 EURUSD
```

learning_rate = 0.001, WINDOW 100 (10 years data - 5m)

```
Epoch 1/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 200s 14ms/step - loss: 0.0024 - val_loss: 1.8054e-05
Epoch 2/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 202s 14ms/step - loss: 2.0486e-04 - val_loss: 2.3588e-05
Epoch 3/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 202s 14ms/step - loss: 1.9804e-04 - val_loss: 5.1064e-05
Epoch 4/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 202s 14ms/step - loss: 1.9222e-04 - val_loss: 1.3759e-05
Epoch 5/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 198s 14ms/step - loss: 1.8954e-04 - val_loss: 2.3596e-05
Epoch 6/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 182s 12ms/step - loss: 1.8737e-04 - val_loss: 1.3825e-05
Epoch 7/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 195s 14ms/step - loss: 1.8312e-04 - val_loss: 2.7451e-05
Epoch 8/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 200s 14ms/step - loss: 1.8290e-04 - val_loss: 7.8285e-05
Epoch 9/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 201s 14ms/step - loss: 1.8079e-04 - val_loss: 1.9231e-05
Epoch 10/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 205s 14ms/step - loss: 1.7950e-04 - val_loss: 1.4362e-05
Epoch 11/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 199s 14ms/step - loss: 1.7898e-04 - val_loss: 9.7222e-06
Epoch 12/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 204s 14ms/step - loss: 1.7399e-04 - val_loss: 1.2207e-05
Epoch 13/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 202s 14ms/step - loss: 1.7534e-04 - val_loss: 1.0713e-05
Epoch 14/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 201s 14ms/step - loss: 1.7348e-04 - val_loss: 1.2930e-05
Epoch 15/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 201s 14ms/step - loss: 1.7233e-04 - val_loss: 9.4828e-06
Epoch 16/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 205s 14ms/step - loss: 1.7162e-04 - val_loss: 2.1516e-05
```

```
Epoch 17/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 183s 13ms/step - loss:
1.7143e-04 - val_loss: 1.5189e-05
Epoch 18/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 198s 14ms/step - loss:
1.7098e-04 - val_loss: 2.6414e-05
Epoch 19/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 198s 14ms/step - loss:
1.7070e-04 - val_loss: 9.1890e-06
Epoch 20/20
14028/14028 ━━━━━━━━━━━━━━━━━━━━ 205s 14ms/step - loss:
1.7132e-04 - val_loss: 8.8409e-06
Restoring model weights from the end of the best epoch: 20.

14028/14028 ━━━━━━━━━━━━━━━━━━━━ 73s 5ms/step - loss:
7.0140e-06
MSE Train: 6.555640538863372e-06
4674/4674 ━━━━━━━━━━━━━━━━━━━━ 24s 5ms/step - loss:
5.7071e-06
MSE Test: 8.840943337418139e-06
4674/4674 ━━━━━━━━━━━━━━━━━━━━ 24s 5ms/step - loss:
9.4284e-06
MSE Val: 7.140496109059313e-06

Test Set RMSE (Original Scale): 0.000412 EURUSD
Test Set MAE (Original Scale): 0.000281 EURUSD
Validation Set RMSE (Original Scale): 0.000370 EURUSD
Validation Set MAE (Original Scale): 0.000293 EURUSD
Training Set RMSE (Original Scale): 0.000355 EURUSD
Training Set MAE (Original Scale): 0.000260 EURUSD
```

learning_rate = 0.001, WINDOW 30 (10 years data - 5m)

```
Epoch 1/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 122s 8ms/step - loss:
0.0025 - val_loss: 5.6063e-05
Epoch 2/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 114s 8ms/step - loss:
2.0617e-04 - val_loss: 2.3681e-05
Epoch 3/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 137s 8ms/step - loss:
1.9872e-04 - val_loss: 1.4632e-05
Epoch 4/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 116s 8ms/step - loss:
1.9538e-04 - val_loss: 1.5186e-05
Epoch 5/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 137s 8ms/step - loss:
1.9257e-04 - val_loss: 3.6138e-05
Epoch 6/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 145s 8ms/step - loss:
1.9058e-04 - val_loss: 2.1825e-05
Epoch 7/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 137s 8ms/step - loss:
1.8649e-04 - val_loss: 3.6287e-05
Epoch 8/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 143s 8ms/step - loss:
1.8648e-04 - val_loss: 1.6698e-05
Epoch 9/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 147s 8ms/step - loss:
1.8294e-04 - val_loss: 1.4644e-05
Epoch 10/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 135s 8ms/step - loss:
1.8181e-04 - val_loss: 1.1773e-05
Epoch 11/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 142s 8ms/step - loss:
1.8183e-04 - val_loss: 1.2267e-05
Epoch 12/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 149s 8ms/step - loss:
1.8191e-04 - val_loss: 1.0225e-05
Epoch 13/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 141s 8ms/step - loss:
1.7737e-04 - val_loss: 1.5016e-05
Epoch 14/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 137s 8ms/step - loss:
1.7867e-04 - val_loss: 1.9609e-05
Epoch 15/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 141s 8ms/step - loss:
1.7771e-04 - val_loss: 4.4310e-05
Epoch 16/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 148s 8ms/step - loss:
1.7731e-04 - val_loss: 1.3341e-05
```

```
Epoch 17/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 143s 8ms/step - loss:
1.7603e-04 - val_loss: 1.7767e-05
Epoch 18/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 137s 8ms/step - loss:
1.7649e-04 - val_loss: 1.3903e-05
Epoch 19/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 145s 8ms/step - loss:
1.7431e-04 - val_loss: 1.3760e-05
Epoch 20/20
14030/14030 ━━━━━━━━━━━━━━━━━━━━ 142s 8ms/step - loss:
1.7249e-04 - val_loss: 1.4544e-05
Restoring model weights from the end of the best epoch: 12.

14030/14030 ━━━━━━━━━━━━━━━━━━━━ 51s 4ms/step - loss:
6.8508e-06
MSE Train: 5.861421868758043e-06
4677/4677 ━━━━━━━━━━━━━━━━━━━━ 17s 4ms/step - loss:
7.2123e-06
MSE Test: 1.022542255668668e-05
4676/4676 ━━━━━━━━━━━━━━━━━━━━ 17s 4ms/step - loss:
1.8515e-05
MSE Val: 7.454897968273144e-06

Test Set RMSE (Original Scale): 0.000443 EURUSD
Test Set MAE (Original Scale): 0.000275 EURUSD
Validation Set RMSE (Original Scale): 0.000378 EURUSD
Validation Set MAE (Original Scale): 0.000249 EURUSD
Training Set RMSE (Original Scale): 0.000335 EURUSD
Training Set MAE (Original Scale): 0.000224 EURUSD
```

learning_rate = 0.001, WINDOW 30 (10 years data - 1m)

```
Epoch 1/20
69671/69671 ━━━━━━━━━━━━━━━━━━━━ 1974s 28ms/step - loss:
7.0529e-04 - val_loss: 7.3281e-06
Epoch 2/20
69671/69671 ━━━━━━━━━━━━━━━━━━━━ 1994s 28ms/step - loss:
1.8147e-04 - val_loss: 1.2228e-05
Epoch 3/20
69671/69671 ━━━━━━━━━━━━━━━━━━━━ 2047s 29ms/step - loss:
1.7570e-04 - val_loss: 7.1345e-06
Epoch 4/20
69671/69671 ━━━━━━━━━━━━━━━━━━━━ 2033s 29ms/step - loss:
1.7142e-04 - val_loss: 5.8495e-06
Epoch 5/20
```

```
69671/69671 1993s 29ms/step - loss:
1.6967e-04 - val_loss: 5.0643e-06
Epoch 6/20
69671/69671 2013s 29ms/step - loss:
1.6728e-04 - val_loss: 6.5425e-06
Epoch 7/20
69671/69671 2108s 29ms/step - loss:
1.6675e-04 - val_loss: 6.4680e-06
Epoch 8/20
69671/69671 2041s 29ms/step - loss:
1.6544e-04 - val_loss: 8.9791e-06
Epoch 9/20
69671/69671 2096s 30ms/step - loss:
1.6380e-04 - val_loss: 1.0729e-05
Epoch 10/20
69671/69671 2086s 29ms/step - loss:
1.6381e-04 - val_loss: 5.6572e-06
Epoch 11/20
69671/69671 2065s 29ms/step - loss:
1.6334e-04 - val_loss: 1.1858e-05
Epoch 12/20
69671/69671 1997s 29ms/step - loss:
1.6301e-04 - val_loss: 6.0791e-06
Epoch 13/20
69671/69671 2057s 29ms/step - loss:
1.6264e-04 - val_loss: 5.0909e-06
Epoch 14/20
69671/69671 2068s 30ms/step - loss:
1.6237e-04 - val_loss: 9.4298e-06
Epoch 15/20
69671/69671 2078s 30ms/step - loss:
1.6264e-04 - val_loss: 7.0916e-06
Epoch 15: early stopping
Restoring model weights from the end of the best epoch: 5.

69671/69671 607s 9ms/step - loss:
3.4239e-06
MSE Train: 3.3350702324241865e-06
23223/23223 205s 9ms/step - loss:
4.0597e-06
MSE Test: 5.0642825044633355e-06
23223/23223 202s 9ms/step - loss:
9.7098e-06
MSE Val: 4.351546067482559e-06

Test Set RMSE (Original Scale): 0.000313 EURUSD
Test Set MAE (Original Scale): 0.000228 EURUSD
Validation Set RMSE (Original Scale): 0.000290 EURUSD
Validation Set MAE (Original Scale): 0.000222 EURUSD
Training Set RMSE (Original Scale): 0.000254 EURUSD
Training Set MAE (Original Scale): 0.000193 EURUSD
```